DD-T-101H
November 17, 1972
SUPERSEDING
Fed. Spec. DD-T-101G
May 20, 1970

FEDERAL SPECIFICATION

TABLEWARE, GLASS (EXTRACTORS, CITRUS FRUIT JUICE (HAND); SALT AND PEPPER SHAKERS; CRUETS, CONDIMENT; PITCHERS; BOTTLES; TUMBLERS; AND OTHER ITEMS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass tableware.

1.2 Classification. Glass tableware shall be of the following types, items, and styles, as specified (see 6.2).

Type I - Not toughened

Items:
- 1 - Extractor, citrus fruit juice, hand (figure 1)
- 4 - Salt and pepper shaker, 3 ounce (figure 2)
- 6 - Pitcher, sirup, 14-ounce (figure 4)
- 7 - Pitcher, sirup, 20-ounce (figure 5)
- 8 - Cruet, condiment, 6-ounce (figure 6)
  - Style A - With ground glass stopper
  - Style B - With plain glass stopper
- 10 - Bottle, water, 32-ounce (figure 7)
- 11 - Pitcher, water, 64-ounce (figure 8)
- 12 - Glass, Sherbet, 6-ounce (figure 9)
- 13 - Dispenser, sugar, 8-ounce (figure 10)
- 14 - Dispenser, sugar, 12-ounce (figure 11)
- 16 - Salt and pepper shaker, 2-ounce (figure 20)
- 17 - Bottle, water, 48-ounce (figure 21)
- 18 - Bowl, food serving, 18-ounce (figure 22)
- 19 - Cruet, condiment, 6-ounce (figure 23)
- 20 - Cup, sherbet, 5-ounce (figure 24)

Type II - Toughened

Items:
- 1 - Tumbler, drinking 6-ounce
  - Style A (figure 12)
  - Style B (figure 18)

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2 - Tumbler, drinking, 10-ounce
Style A (figure 13)
Style B (figure 14)
3 - Tumbler, drinking, 12-ounce (figure 15)
4 - Tumbler, drinking, 4-ounce (figure 16)

Type III - Lead semi-crystal

Items: 1 - Bottle, water (figure 25)
2 - Bowl, food serving (figure 26)
3 - Cruet, condiment
Style A (figure 27)
Style B (figure 28)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

P-S-620	-	Soap, Toilet, Cake (Floating and Nonfloating).
QQ-S-766	-	Steel Plates, Sheets and Strips - Corrosion Resisting.
UU-P-553	-	Paper, Wrapping, Tissue.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR 121.2501:

Food Additives Resulting From Contact With Containers or Equipment and Food Additivies Otherwise Affecting Food; Olefin polymers.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is ifentified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass.

3.1.1.1 Type I, not toughened. The ware shall be made from either a lime, borosilicate or a lead-flint glass, and may be either handmade or machine-made.

3.1.1.2 Type II, toughened. The ware shall be as specified for type I, and shall be heat-treated to increase its resistance to mechanical shock.

3.1.1.3 Type III, lead semi-crystal. The ware shall be made from a lead-silica-alkali glass, commonly known as lead semi-crystal. The lead-silica-alkali glass shall contain not less than 9-1/2 percent lead oxide. Type III ware shall be either hand made and hand polished or shall be manufactured by the hand-operated mold-blown process at the option of the manufacturer. Ware manufactured by the hand-operated mold-blown process shall be finished by hand-cutting and hand-polishing. Vissible parting lines that are inherent in the hand-operated mold-blown process shall not be cause for rejection.

3.1.2 Fittings.

3.1.2.1 Salt and pepper shakers. Fittings for the salt and pepper shakers shall be made of corrosion-resisting steel conforming to class 302 or 430 of QQ-S-766. The thickness of the steel used to make the fittings shall be a minimum of 0.009 inch.

3.1.2.2 Sirup pitchers. Fittings for the sirup pitchers shall be made of plastic and metal, or corrosion-resisting steel conforming to class 302 or 430 of QQ-S-766, as specified (see 6.2).

3.1.2.3 Sugar dispensers. The caps for the sugar dispensers shall be made of corrosion-resisting steel conforming to class 302 or 430 of QQ-S-766.

3.1.2.4 Toxicity. All plastic parts shall conform to 21 CFR 121.2501.

3.1.2.5 Odor. All plastic parts shall be free from objectionable odor when tested as specified in 4.3.2.3.

3.2 Design, construction and shape. The tableware shall conform to the designs and shapes shown on figures 1 through 28 of this specification as applicable, with dimensional tolerances as indicated herein and on the figures.

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3.2.1 Ware (glass). The ware shall be constructed of a quality to be free of heavy cords, checks, soft blisters and open blisters. Stones shall have diameters not greater than 0.03 inch if well buried, or 0.02 inch if near the surface. The ware may have a few scattered bubbles up to 0.05 inch in diameter and elongated bubbles up to 0.10 inch in length. Type I and II glass may be slightly colored and type III glass shall be colorless.

3.2.1.1 Shape. The ware shall be straight and true so when rotated about an axis perpendicular to and passing through the center of the base, the top edges shall not show an eccentricity greater than 0.12 inch. The difference between the major and minor axis of out-of-round ware shall not exceed 2 percent of the specified diameter, where applicable. The ware shall not rock more than 0.008 inch for types I and II nor more than 0.020 inch for type III when placed on a flat surface.

3.2.2 Plastic. All plastic parts shall conform to the shape and dimensions shown in figure 4. Threaded fittings shall be attached finger tight to the ware.

3.2.3 Sugar dispensers. The caps for the sugar dispensers shall be side-opening covered by a corrosion resisting steel flap that covers the pouring opening when the dispenser is in upright position (see figures 10 and 11).

3.3 Performance.

3.3.1 Resistance to boiling. The glassware shall show no evidence of corrosion, scumming, chipping, or cracking when tested as specified in 4.3.1.1.

3.3.2 Thermal shock resistance. The glassware shall show no evidence of cracking or checking when tested as specified in 4.3.1.2.

3.3.3 Type II toughened ware.

3.3.3.1 Impact strength. The toughened glassware shall have an impact strength of not less than 0.016 foot-pounds at the outside rim and 0.005 foot-pounds at the inside rim when tested as specified in 4.3.1.3.1 and 4.3.1.3.2.

3.3.3.2 Toughness. Breakage of glassware shall be nondicing in character, and not more than 15 pieces (excluding slivers and chips less than 1/4 inch long) shall occur when tested as specified in 4.3.1.4.

3.3.3.3 Abrasion. The glassware shall not rupture, crack, or chip when tested as specified in 4.3.1.5.

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3.4 Plastic stability (boiling and washing). No plastic part shall change dimensions sufficiently to interfere with its proper fit on the glass containers, and there shall be no evidence of chalking, cracking or softening when tested as specified in 4.3.2.1 and 4.3.2.2, respectively.

3.5 Finish.

3.5.1 Glass. All surfaces and edges shall be smoothly finished. Edges of tumblers may be either unbeaded or may have beads up to 0.12 inch in thickness.

3.5.2 Metal. All metal parts shall be properly ground to remove all surface imperfections, polished with emery, and finished with an iron-free abrasive, rouge or polish, to remove all emery grind marks and shall result in a well-polished finish.

3.5.2.1 Salt and pepper shakers. The metal caps shall have a burnished finish.

3.5.3 Plastic. The surfaces of all plastic parts shall be free of mars and blemishes. All edges shall be properly buffed to remove roughness. Holes shall be completely cleaned of flash or fins.

3.6 Marking. Tableware shall be permanently and distinctly marked in accordance with the manufacturer's commercial practice.

3.7 Workmanship. The tableware shall be uniformly constructed and free from cracks, chips, blisters, or large stones (see 3.2.1). The edges shall be well rounded and sufficiently fire polished to remove fins and sharp edges. All threaded members shall fit their respective ware and shall be capable of being attached to, and detached from, the respective ware by normal hand pressure. All moving parts of the sirup cap shall move freely and smoothly. The glass stoppers shall fit snugly in the cruets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check-test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects set forth in the applicable sub-paragraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.2.1.4. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of complete items of tableware of one type, item, and style for examination in 4.2.2.1.1 and 4.2.2.1.2 and in units of shipping containers for examination in 4.2.2.1.3.

4.2.2.1.1 Visual examination. The sample unit for this examination shall be one complete item of tableware. Examination shall be made for the defects in table I.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Appearance	Not clean; any marked discoloration of an individual item	X	
	Not colorless (type III only)	X	
	Feathering	X	
	Any mottled areas, visible particles of foreign or unmelted material	X	
Workmanship	Not type, item, or style specified	X	
	Nick, chip, fracture, or crack (including firecracks)	X	

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TABLE I. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Workmanship (cont'd)	Waves, cords or shear marks	X	
	Open or soft blisters	X	
	Scratches or indentations		X
	Stones greater than 0.03 inch if well buried or greater than 0.02 inch if near surface	X	
	Round bubble greater than 0.05 inch in diameter	X	
	Elongated bubble greater than 0.10 inch in length	X	
	Any defect affecting appearance or serviceability	X	
Construction Glassware	Design varies from style and shape shown in applicable figure	X	
	Corners or edges not well rounded or not blended smoothly (except as specified in 3.1.1.3).	X	
	Item of tableware does not rest evenly and smoothly when placed in normal position on a flat surface (rock greater than 0.008 inch for types I and II or 0.020 inch for type III).	X	
	Identification marking not as specified.	X	
Fittings	Salt and pepper fittings not made of corrosion-resisting steel	X	
	Sirup fittings not made of plastic and metal or corrosion-resisting steel	X	
	Sirup fittings do not move freely and smoothly	X	
	Sugar dispenser fitting not made of corrosion resisting steel	X	
	Any sharp ends or protrusions	X	
	Metal parts not properly ground and polished; not well-polished finish		X
	Fittings twisted, bent, or deformed, do not fit glassware properly	X	
	Openings not completely cleaned of flash or fins		X

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TABLE I. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Fittings (con.)	Plastic parts not free of mars or blemishes		X
	Cracks or fractures in metal or plastic parts	X	
	Any part missing from an assembly	X	
	Component members do not fit snugly	X	
	Threaded members cannot be attached or detached from respective ware by normal hand pressure	X	

4.2.2.1.2 Examination of the end item for defects in capacity, weight, and dimensions. The sample unit for this examination shall be one complete item of tableware.

Examine	Defect
Capacity	Less than the minimum capacity indicated in the applicable figure for the specified type, item, and style tableware, when applicable
Weight	Any weight deviation greater than the tolerance permitted in the applicable figure for the specified type, item, and style tableware, when applicable
Dimensions	Any dimensional deviation greater than the tolerances permitted by the applicable figure for the specified type, item, and style of tableware

4.2.2.1.3 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing and marking complies with the section 5 requirements. The sample unit shall be one shipping container fully prepared for delivery except that it shall not be closed. Closed shipping containers shall be examined for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot.

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Examine	Defect
Markings (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application
Material	Any nonconforming component; component missing, damaged, or otherwise defective
Workmanship	Inadequate application of components, such as incomplete closure of container flaps; loose strapping, inadequate stapling, or improper taping Bulged or distorted container
Contents	More or less than the specified number of items per container

4.2.2.1.4 Inspection levels and AQL's for examinations. The inspection levels, for purposes of determining the sample size in accordance with MIL-STD-105, and the AQL's expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL	
		Major	Total
4.2.2.1.1	I	2.5	4.0
4.2.2.1.2	S-3	One class	4.0
4.2.2.1.3	S-1	One class	2.5

4.2.2.1.5 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified

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Examine	Defect
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.2.3 Testing of the end item. The end item shall be tested for the applicable characteristics in table II. The lot size shall be expressed in units of complete items of tableware of one type, item, and style offered for inspection at one time. The sample unit shall be one complete item of tableware. The sample size shall be drawn at inspection level S-2 of MIL-STD-105. The failure of any sample unit to pass any applicable test shall be cause for rejection of the lot.

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TABLE II. Testing of the end item

Characteristic	Specification reference: Requirement paragraph	Specification reference: Test	Requirements applicable to Individual unit	Number of determinations per sample unit	Results reported as pass or fail 1/
Glassware					
All types					
Composition	3.1.1.1 3.1.1.2 or 3.1.1.3	2/			
Boiling resistance	3.3.1	4.3.1.1	X	1	X
Thermal resistance	3.3.2	4.3.1.2	X	1	X
Type II only					
Impact strength					
Outside rim	3.3.3.1	4.3.1.3.1	X	1	X
Inside rim	3.3.3.1	4.3.1.3.2	X	1	X
Toughness	3.3.3.2	4.3.1.4	X	1	X
Abrasion	3.3.3.3	4.3.1.5	X	1	X
Plastic parts					
Stability (boiling)	3.4	4.3.2.1	X	1	X
Stability (washing)	3.4	4.3.2.2	X	1	X
Odor test	3.1.2.5	4.3.2.3	X	1	X

1/ If failure is indicated, report description of failure.

2/ When specified (see 6.2) the supplier shall furnish a certificate of compliance stating that the glass conforms to the requirements of 3.1.1.1, 3.1.1.2, or 3.1.1.3, as applicable.

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4.3 Test procedures.

4.3.1 Glass parts.

4.3.1.1 Boiling test. Samples shall be completely immersed in boiling distilled water for 6 hours in a closed vessel having a small steam vent, boiling water being occasionally added to compensate for evaporating losses. The samples shall be suspended in the water so they do not come in contact with the containing vessel. Any sample which shows corrosion, scumming, chipping, or cracking as a result of the boiling test shall be considered a failure.

4.3.1.2 Thermal shock test. The samples shall be tested by immersing them in a hot water bath having a temperature as indicated in table III, allowing them to remain for 5 minutes, then transferring them and immediately immersing them for 30 seconds in a cold water bath having a temperature of 70° ± 2°F. The 5 minute immersion in the hot bath shall be controlled within 3 seconds, and the time taken for transfer from the hot to the cold bath shall be 15 ± .33 seconds for type I and II glass and 20 to 21 seconds for type III glass. The water shall not be emptied from the samples during the transfer from the hot to the cold bath. During the test the samples shall be protected from drafts. Any sample that cracks, chips, or checks during, or as a result of, the thermal shock test shall be considered a failure.

TABLE III. Temperature of the hot bath

Type I items	Degrees F. + 1°
Item 1-Extractor citrus fruit juice hand	140
Items 4 and 16-Salt and pepper shakers, 3-ounce and 2-ounce	145
Item 6-Pitcher, sirup 14-ounce	145
Item 7-Pitcher, sirup 20-ounce	140
Items 8 and 19-Cruets, condiment, 6-ounce	145
Item 10-Bottle, water, 32-ounce	145
Item 11-Pitcher, water, 64-ounce	140
Item 12-Glass, Sherbet, 6-ounce	160
Item 13-Dispenser, sugar, 8-ounce	145
Item 14-Dispenser, sugar, 12-ounce	145
Item 17-Bottle, water, 48-ounce	145
Item 18-Bowl; food serving, 18-ounce	160
Item 20-Cup, sherbet, 5-ounce	140

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TABLE III. Temperature of the hot bath (con.)

Type I items	Degrees F. ± 1°
Type II items	
All items	160
Type III items	
All items	140

4.3.1.3 Impact strength (type II).

4.3.1.3.1 Outside-rim impact test (type II only). Before making the outside rim impact test (for type II only), the inside surface of the rim shall be abraded parallel to the sample's principal axis (one revolution) by means of fresh No. 150 grit silicon carbide paper or cloth mounted on a steel cylinder 1 inch in diameter, and 4 inches long. The method used for abrading the ware shall be as follows: The sample and the cylinder shall both be held in approximately horizontal position, and the cylinder, lightly held at one end, shall be permitted to rest the major portion of its weight on the rim of the sample. The cylinder shall then be moved briskly in and out, and slowly rotated as the sample also is very slowly rotated. The abraded sample shall be placed bottom downward on a flat steel plate, and sample wall shall be supported at the bulge, or if not bulged at a point 1 inch below the rim against a V-shaped hard fiber support which is 3 inches wide, 1-inch high, 1-inch thick at the ends, and 1/2-inch thick at the base of the "V". The fiber support shall have a hardness of 85 plus or minus 10 of the Rockwell R scale, determined as described in Fed. Test Method Std. No. 406. The fiber support shall be backed up by a 0.25 inch steel plate. Impact shall be made on the outside surface of the rim by a pendulum hammer swinging in the plane of the angle bisector of the V-block. The effective pendulum rod length shall be 3 inches, pendulum rod diameter 3/32-inch, and hammer made of a 1-inch steel ball on which equal parts of the top and bottom have been ground off flat, making it 1/2-inch high. The striking surface of the pendulum bob shall have a hardness of 20 or more on the Rockwell C scale. The apparatus shall be adjusted so that when the pendulum hangs vertically, its point of impact lightly touches the outside of the rim of the sample at a point opposite the fiber support. Impact shall be made once at each of four points on the rim, approximately 90° apart, with the energy at a constant value of 0.016 foot-pound. Any sample that cracks, chips, checks, or breaks shall be considered to have failed the test.

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4.3.1.3.2 Inside-rim impact test (type II only). Procedure and equipment for the inside-rim impact test (for type II only) shall be the same as that used for the outside-rim impact test described above, except that (1) the outside surface of the rim only shall be abraded, (2) impact shall be made as above on the inside surface of the rim at the point nearest the vertex or the angle of the V-block, and (3) impact energy shall be at a constant value of 0.005 foot-pound. Any sample that cracks, chips, checks, or breaks shall be considered to have failed the test.

4.3.1.4 Breakage test (type II only). The breakage test shall be performed using a machine identical with that used for the impact tests, except that a larger pendulum shall be used. The effective pendulum rod length shall be 5 inches, pendulum rod diameter 3/32-inch, and hammer made of 1-1/4 inch steel ball on which equal parts of the top and bottom have been ground off flat, making it 1/2-inch high. The striking surface of the pendulum bob shall have a hardness of 20 or more on the Rockwell C scale. The inside surface of the sample shall be abraded as specified in 4.3.1.3.1. The sample shall be supported in the V-shaped support, and impact shall be made on the outside of its rim by releasing the pendulum from the 90° position. If necessary, the blow shall be repeated until breakage occurs. Any breakage dicing in character and more than 15 pieces, excluding slivers and chips less than 1/4-inch long shall be considered to have failed the test.

4.3.1.5 Abrasion test (type II only). Each sample shall be thoroughly abraded on the entire inner and outer surface using fresh No. 50 grit silicon carbide paper or cloth. After abrading, each sample shall be immersed in tap water for 5 minutes. The development of a rupture, or any cracking or chipping shall constitute failure.

4.3.2 Plastic tests.

4.3.2.1 Boiling test. The plastic shall be immersed in boiling distilled water for 30 minutes, removed, and immediately immersed in water at room temperature (68° to 78°F.) for 5 minutes. Upon removal from this immersion, the parts shall be allowed to air-dry at room temperature for 4 hours. Any part showing dimensional changes sufficient to interfere with its proper fit on the glass containers, or evidence of chalking, cracking, or softening as a result of the test shall be considered as having failed the test.

4.3.2.2 Washing test. The plastic parts shall be immersed for 30 minutes in water at 70° Centigrade (160°F.) which contains 0.1 percent soap conforming to type I, class 1 of P-S-620 and 0.1 percent washing soda (Na_2CO_3, $10H_2O$).

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Upon removal from the solution, the parts shall be allowed to air-dry at room temperature for 1 hour. This cycle shall be immediately repeated to give a total of five such cycles. After the fifth cycle, the parts shall be rinsed with distilled water to remove soap and alkali, and allowed to air-dry before examination. Any part showing dimensional changes sufficient to interfere with its proper fit on the glass container or evidence of chalking, cracking, or softening as a result of the test shall be considered as having failed the test.

4.3.2.3 Odor test for plastic parts. Place the plastic parts of the sample units to be tested in a clean, dry glass container approximately 4 to 6 times the volume of the items to be tested. Close the container, place in a circulating air oven, maintaining a temperature of 125° ± 5°F. for 4 ± 1/4 hours. The container shall then be removed from the oven and opened, and immediately checked for odor. The presence of a strong odor or disagreeable chemical odor shall be considered as having failed the test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each shipping container shall have inserted sets of slotted partitions so designed as to provide cells for the items. Dividing pads, when applicable, shall be used to separate the sets of slotted partitions. Where more than one item is placed in an individual cell, the items shall be separated from one another by indented paper pads. Cruet stoppers shall be individually wrapped in at least two thicknesses of tissue paper conforming to type I, class 1 or 2 of UU-P-553, or placed in an end-opening style kraft paper envelope, and the flap securely sealed to prevent escape of contents. Cruet stoppers, prepared as specified, shall be placed together with cruets within the individual cells. The inside of each PPP-B-636 shipping container shall be fitted with a taped one-piece box liner fabricated as specified in 5.1.1.1.

5.1.1.1 Partitions, pads, and liners. Each slotted partition, dividing pad, and taped box liner shall be fabricated from fiberboard material conforming to type CF (A or C flute) class domestic, variety SW, grade 200 of PPP-F-320.

5.1.2 Level C. Tableware shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Tableware, of one description only, packaged as specified in 5.1, shall be packed in the quantities specified in table IV within a snug-fitting fiberboard shipping container conforming to style CSSC-L, grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636.

TABLE IV. Quantity of tableware per shipping container (see 6.2)

Tableware items	Quantity (each) per shipping container
Extractor, citrus fruit juice, hand	24
Salt and pepper shaker	72 or 144
Pitcher, sirup	12 or 24
Cruet, condiment	12 or 24
Bottle, water	6, 12 or 24
Pitcher, water	6 or 12
Cups or glasses, sherbet	72
Tumblers, drinking	72
Dispenser, sugar	24
Bowl, food serving	12

5.2.2 Level B. Tableware of one description only, packaged as specified in 5.1, shall be packed in the quantities specified in table IV within a snug-fitting fiberboard shipping container conforming to style CSSC-L, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C. Tableware, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization. (Military requirements). Unless otherwise specified (see 6.2), tableware of one description packed as specified in 5.2, shall

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be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123. Each shipping container shall have printed or stenciled on top of the container in capital letters not less than 3/4-inch in height the following:

GLASS - DO NOT DROP OR THROW

5.4.1.1 Fragile marking. At least three surfaces of each shipping container shall bear a white paper stock label, with a 1/4-inch red border, fracture features, and the word "FRAGILE". On containers for all tableware items, the fragile label shall be 4-inches by 4-inches, and shall be so placed as to be conspicuous, but not to interfere with other markings. The label shall not be placed on the lower one-third of the sides of the container, unless specifically required by law.

5.4.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.4.2.1 Special marking. Each shipping container shall have fragile marking applied in accordance with MIL-STD-129. In addition, the top of each container shall be plainly marked in capital letters not less than 3/4-inch in height as follows:

GLASS
HANDLE WITH CARE

6. NOTES

6.1 The glass tableware is intended for use by civil and military agencies in serving, dispensing, and storing food, water, or condiments as applicable.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, item and style required (see 1.2).

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(c) Whether fittings for sirup pitchers shall be made of plastic and metal or corrosion-resisting steel (see 3.1.2.2).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) Packed quantity of tableware required where a multiple choice exists (see table IV).
(f) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(g) When palletization is not required for military use (see 5.3).
(h) When a certificate of compliance for glass is required (see table II).

6.3 The following items specified in DD-T-101G have been deleted from this revision of DD-T-101.

Type I	Type II	Type III
Item 2	Item 5	Item 4
Item 3	Item 6	
Item 5		
Item 15		

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 84

Review activities:

Army - MD, ME
Navy - MC, MS

User activity:

Army - SM

Preparing activity:

Army - GL

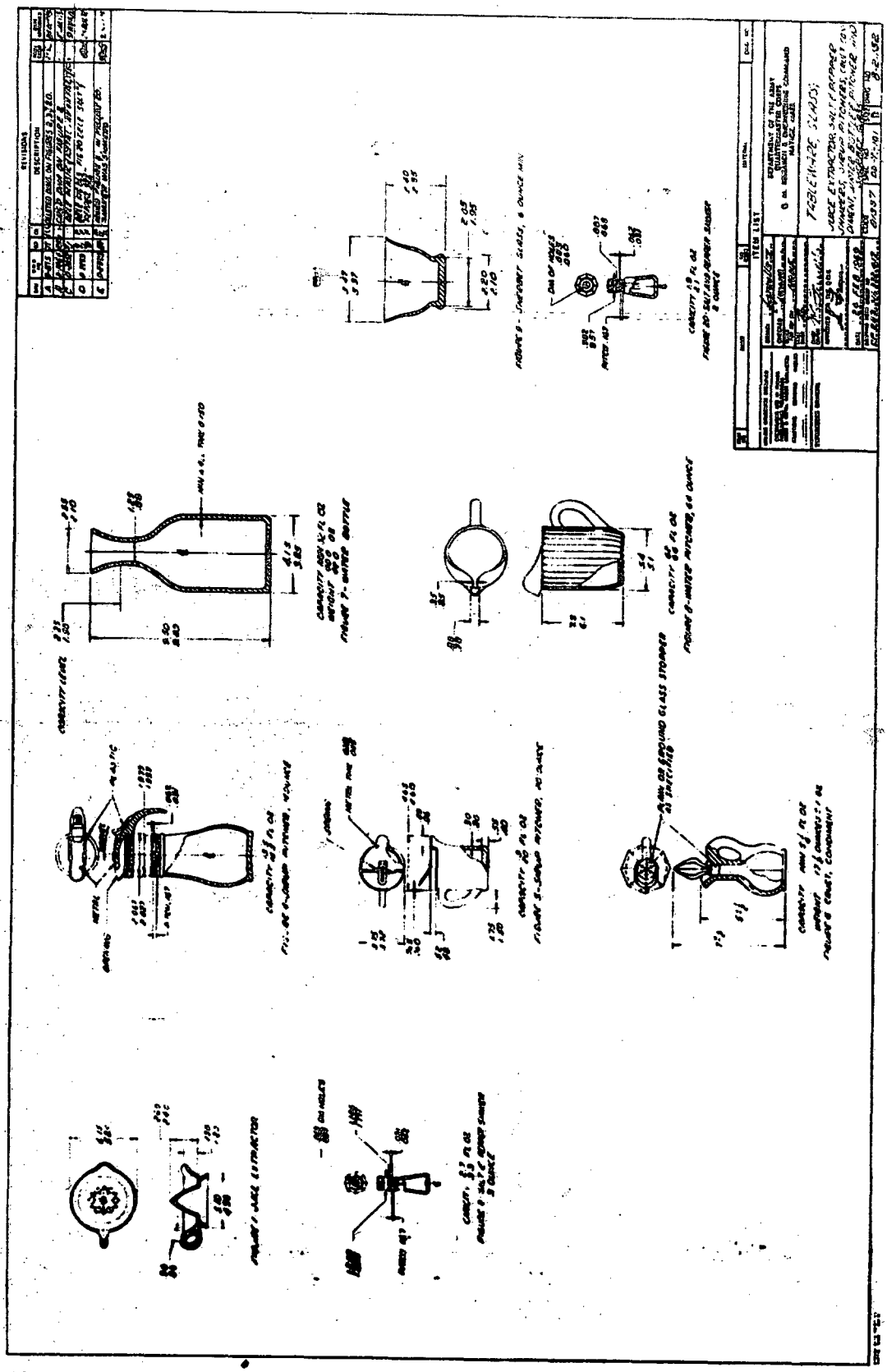
Civil Agency Coordinating Activities:

GSA-FSS
USDA-AFS
VA-DMS

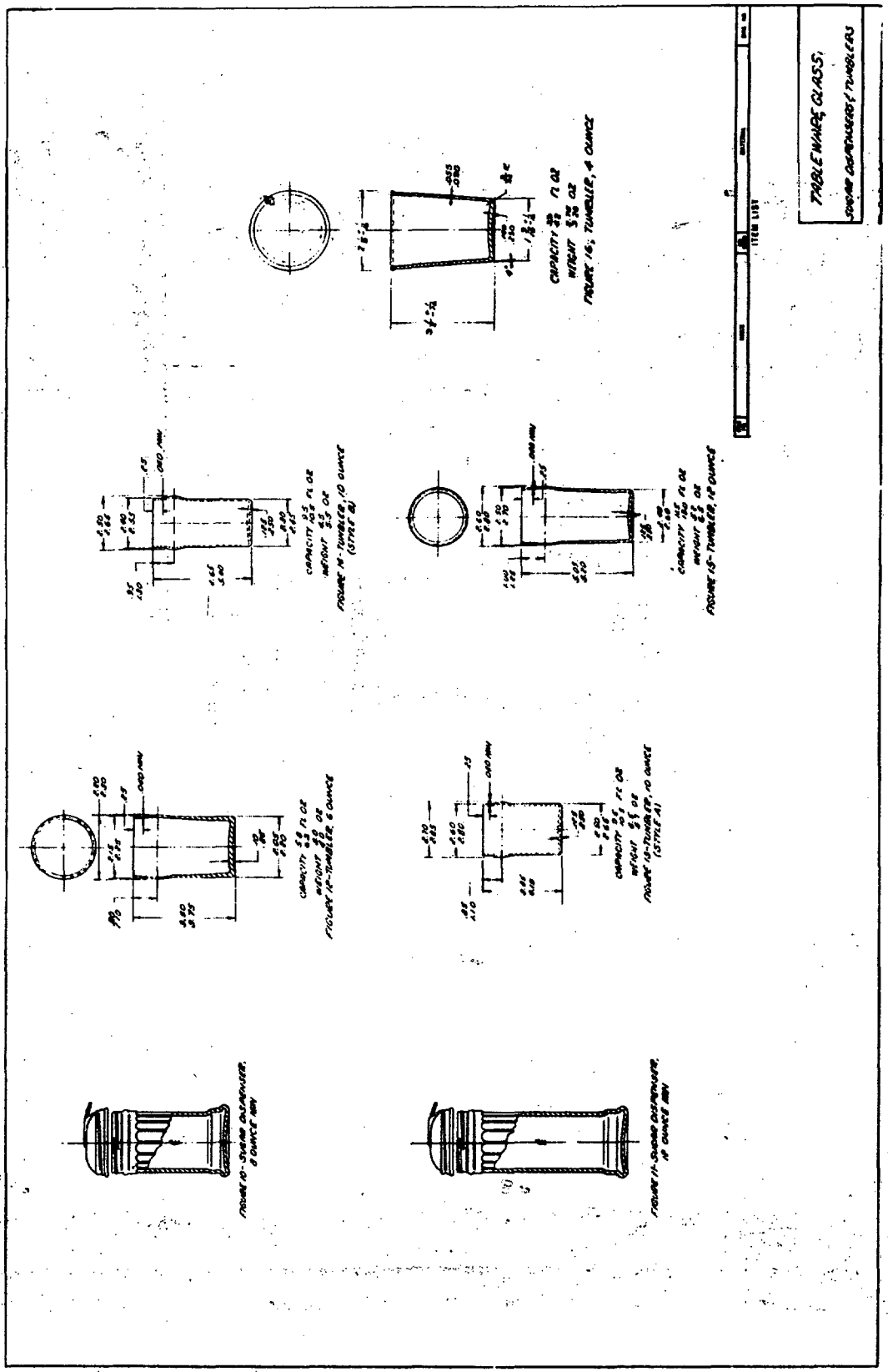
Project No. 7350-0148

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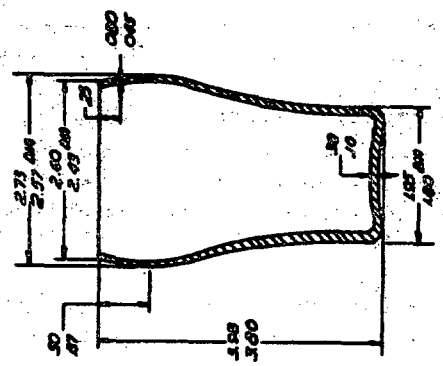
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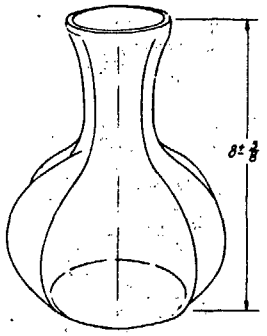
TABLEWARE, GLASS:
TUMBLERS

ITEM LIST

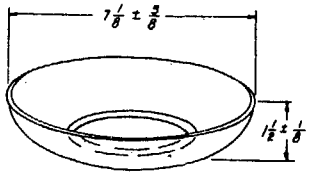
PLATE 19-TUMBLER, 6OZ

CAPACITY 6 FLOZ
HEIGHT 3.25
DIA 2.75

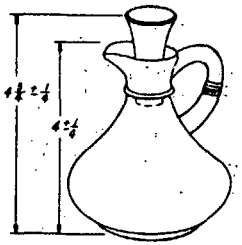

DIA
DIA
DIA



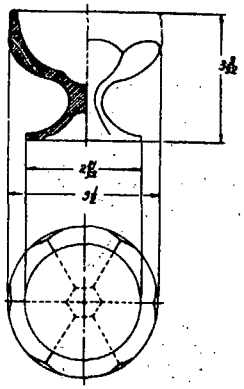
CAPACITY MIN 50 FL OZ
WEIGHT MIN 17 OZ
FIGURE 21- WATER BOTTLE
48 OUNCE



CAPACITY MIN 19 FL OZ
WEIGHT MIN 15 OZ
FIGURE 22- BOWL FOOD SERVING
18 OUNCE



CAPACITY MIN 6 1/2 FL OZ
WEIGHT MIN 7 1/2 OZ
FIGURE 23- CREAM CONDIMENT
6 OUNCE



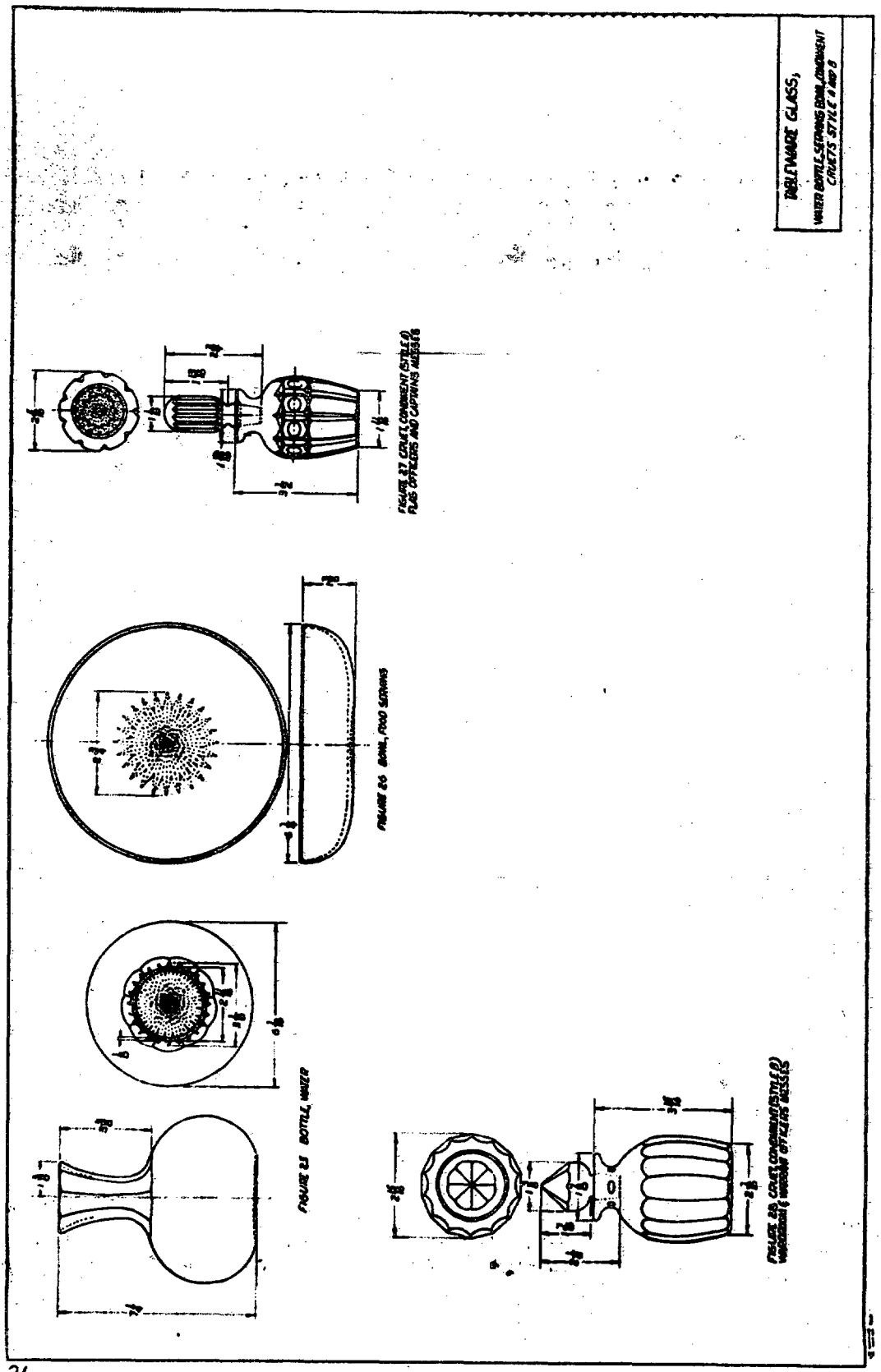
CAPACITY APPROXIMATELY 5 OZ
TOLERANCE 1/4 INCH
FIGURE 24- SHERBET CUP

TABLEWARE GLASS;

WATER BOTTLE, SERVING BOWL, CONDIMENT CREAM, AND SHERBET CUP

DD-T-101H

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